Archives of Microbiology (2022) 204:87
https://doi.org/10.1007/s00203-021-02675-y

ORIGINAL PAPER

Metagenomic landscape of taxonomy, metabolic potential and resistome of *Sardinella longiceps* gut microbiome

Tina Kollannoor Johny[1] · Rinu Madhu Puthusseri[1] · Sarita Ganapathy Bhat[1]

Received: 1 July 2021 / Revised: 27 November 2021 / Accepted: 7 December 2021 / Published online: 27 December 2021

Abstract

Fish gut microbiota, encompassing a colossal reserve of microbes represents a dynamic ecosystem, influenced by a myriad of environmental and host factors. The current study presents a comprehensive insight into *Sardinella longiceps* gut microbiome using whole metagenome shotgun sequencing. Taxonomic profiling identified the predominance of phylum *Proteobacteria*, comprising of *Photobacterium*, *Vibrio* and *Shewanella* sp. Functional annotation revealed the dominance of Clustering based subsystems, Carbohydrate, and Amino acids and derivatives. Analysis of Virulence, disease and defense subsystem identified genes conferring resistance to antibiotics and toxic compounds, like multidrug resistance efflux pumps and resistance genes for fluoroquinolones and heavy metals like cobalt, zinc, cadmium and copper. The presence of overlapping genetic mechanisms of resistance to antibiotics and heavy metals, like the efflux pumps is a serious cause of concern as it is likely to aggravate co-selection pressure, leading to an increased dissemination of these resistance genes to fish and humans.

Keywords Metagenomics · *Sardinella longiceps* · Taxonomy · Antibiotic resistance · Heavy metal resistance

Introduction

The gastrointestinal tract of fishes is colonized by complex microbial communities that encompass bacteria, archaea, viruses and fungi. They reinforce the metabolic functionalities of the host in respect of digestion of nutrients, energy harvesting, defense against pathogens and maturation of the immune system (Sullam et al. 2012). The composition and function of the microbiota are influenced by host factors such as diet, genetics, phylogeny, gender, age and habitat factors such as temperature, salinity, presence of pollutants/toxins and specific rearing conditions in the case of aqua cultured fish species (Ghanbari et al. 2015).

Culture-dependent techniques were the mainstay of fish gut microbiota studies during the past decades. However, the emergence of culture-independent techniques such as quantitative real-time PCR (qPCR), fingerprinting techniques and fluorescent in situ hybridization (FISH) led to a dramatic expansion of knowledge on the community composition of fish gut microbiota (Zhou et al. 2014). Recently, metagenomic approach has become the method of choice for such investigations as they elucidate the community composition as well as functional attributes of complex environmental samples. This aids to establish links between the host, gut microbiota, and the ecosystem. Metagenomic approach includes both sequence-based and function-based analyses of microbial community DNA (Riesenfeld et al. 2004).

The recent advancements in next-generation sequencing techniques leading to a decline in the sequencing costs and an improvement of speed, depth and accuracy has driven the focus to mostly sequence based metagenomic approaches during the past few years (Foster et al. 2012; Ghanbari et al. 2015). Among the sequence-based approaches, whole metagenome shotgun sequencing permits comprehensive sampling of all the microbial genes present in an environmental sample and generates a high-resolution picture of the taxonomy and functional potential of the microbiome (Tringe et al. 2005). Apart from providing a great resolution of taxonomy of bacteria, archaea, viruses, and eukaryota comprising the microbiome, shotgun sequencing also facilitates the discovery of novel genes (Shah et al. 2011; Johny et al. 2021).

Communicated by Erko Stackebrandt.

Sarita Ganapathy Bhat
saritagbhat@gmail.com

[1] Department of Biotechnology, Cochin University of Science and Technology, Kalamassery, Cochin, Kerala 682022, India

As fishes constitute an important food source, with over 3 billion of the global population dependent on fishes for 20% of their protein intake, it is important to consider them in the expanding discipline of host microbial research (Egerton et al. 2018). However, most research on fish gut microbiota was undertaken to study the effect of various host and environmental factors on the composition of fish gut microbiota. The findings from these studies were directed to fortify or manipulate the gut microbiota and improve fish health and aquaculture productivity (Sullam et al. 2012; Ringø et al. 2016; Liu et al. 2016).

A recent investigation of Atlantic cod gut microbiome revealed that the gut microbiome serves as a biomarker of oil contamination in the marine environment (Walter et al. 2019). The marine environment is also a genetic reservoir of antibiotic resistance owing to anthropogenic runoffs of antibiotics and/or antibiotic resistant bacteria and the selection of resistant bacteria in response to antagonistic microbial interactions (Hatosy and Martiny 2015). Only a handful of studies have investigated the prevalence of antibiotic resistance genes in fish gut microbiomes using metagenomic approach (Tyagi et al. 2019; Sáenz et al. 2019; Khurana et al. 2020). Heavy metal pollution is another rampant issue of the marine ecosystem that leads to bioaccumulation and biomagnification of toxic metals in marine food webs, including fish (Olayinka-Olagunju et al. 2021).

The Indian oil sardine, *Sardinella longiceps* is a pelagic clupeid fish, distributed in the northern and western parts of the Indian Ocean. It is a valuable commercial fish that constitutes a substantial proportion of the marine fish catch of India. On an average, India contributed to around 80% of the global oil sardine catch from 1950 to 2014 (Kripa et al. 2018). Apart from its food value, the fish also finds application in the production of fish meal and oil. Till date, only a handful of studies have explored the gut bacteria of *S. longiceps*. However, these studies were focussed on the characterization of a few bacterial isolates with bioactive potential (Augustine and Joseph 2018; Saidumohamed and Bhat 2021; Saidumohamed et al. 2021; Susmitha et al. 2017) Since whole metagenome shotgun sequencing helps to obtain single snapshot view of all the plausible host microbial interactions in a specific environment, the present study employed shotgun metagenome sequencing to discern the taxonomic and metabolic profiles of *S. longiceps* gut. The sequencing data was also subjected to characterization of the resistome profiles for antibiotics and toxic compounds, including heavy metals.

Materials and methods

Metagenomic DNA extraction and sequencing

Healthy adult *S. longiceps* ($n = 5$; BL $= 16.28 \pm 0.92$ cm; BW $= 125 \pm 1.12$ g) of ring seine catch from 50 m depth (9° 85′ N and 76° 11′ E) were freshly collected from Thoppumpadi harbour (Kochi, Kerala, India) during the morning hours (IST 6:00 A.M.–7:00 A.M.) in January 2019. The gut contents of the fishes were excised under aseptic conditions and pooled for the isolation of metagenomic DNA. Metagenomic DNA was extracted from the gut contents using a previously optimized protocol in our lab (Tina et al. 2014) with minor modifications and 178 ng of DNA was taken for library preparation. De novo whole metagenome shotgun sequencing was performed using Illumina HiSeq 2000 paired-end technology (200 cycles) with a targeted paired end read length of 100 bp at AgriGenome Labs Pvt Ltd, Cochin, Kerala, India.

Metagenomic sequencing data analysis

The raw fastq files were quality filtered and adapter trimming was performed using Cutadapt (ver 1.8.1). The paired end sequence files were uploaded to MG-RAST server (metagenomic rapid annotations using subsystems technology, v 4.0.3 (http://www.mg-rast.org/index.html). MG-RAST is an open-source, open-submission platform used to perform taxonomic and functional analyses of environmental datasets. The overlapping paired-end reads were joined and processed using the MG-RAST pipeline.

Taxonomic and functional annotations of the metagenome reads were performed against RefSeq and SEED Subsystems databases, respectively. All the annotations were performed using the default parameters in the MG-RAST pipeline (maximum *e* value of 10^{-5}, a minimum identity of 60% and minimum alignment length of 15 bp). Taxonomic assignment for the bacterial domain was done at phylum and genus levels. The SEED subsystems based functional assignment was retrieved by applying filters at various hierarchical levels of classification to compute relative abundance of genes involved in different metabolic functions. For resistome profiling, the metagenomic dataset were analyzed for reads grouped at Resistance to antibiotics and toxic compounds (level 2) of 'Virulence, Disease and Defense' (level 1) of the SEED subsystems annotation. Level 3 and function level subsystems were also analyzed for a deeper understanding of the resistome profiles.

Results

Metagenomic DNA extraction and sequencing

Whole metagenome shotgun sequencing on Illumina HiSeq 2000 generated 2.64 GB data with 26,451,026 reads with a length of 100 bp. The sequence files were uploaded to MG-RAST server for taxonomic and functional annotation. Out of the sequences that passed the QC pipeline, 76.89% reads mapped to proteins with known functions, 17.89% reads contained proteins with unknown function and 5% reads contained rRNA genes. The sequencing data from the study are available in the NCBI BioSample Submission Portal as BioProject PRJNA433183 and SRA accession number SRR6677342.

Taxonomic profiling of *Sardinella longiceps* gut microbiome

Taxonomic annotation of the reads was performed against the RefSeq database using the Best Hit Classification algorithm of MG-RAST. At the domain level, bacteria accounted for 99.64% of the reads, followed by viruses (0.17%), eukaryota (0.15%) and archaea (0.03%). Further, annotation of the bacterial domain was performed at phylum and genus levels. At the phylum level, *S. longiceps* gut microbiome was dominated by *Proteobacteria* (99.13%), followed by *Firmicutes* (0.27%), *Actinobacteria* (0.15%), *Cyanobacteria* (0.13%), and *Bacteroidetes* (0.11%). The remaining 22 phyla and the unknown phyla accounted for 0.21% of the reads (Fig. 1) (Supplementary Table S1). At the genus level, the major bacterial groups identified were *Photobacterium* (72.67%), *Vibrio* (11.68%), *Shewanella* (3.08%), *Bacillus* (0.09%) and *Synechococcus* (0.07%) (Fig. 2). The remaining 547 genera and unknown reads accounted for 12.41% of the sequencing reads (Supplementary Table S2).

Functional profiling of *Sardinella longiceps* gut microbiome

Functional analysis of the *S. longiceps* gut microbiome was performed by annotation of the whole metagenome sequencing reads against the SEED Subsystems database. Functional analysis at subsystems level 1 classification revealed the predominance of Clustering based subsystems (11.28%) and Carbohydrate subsystems (11.16%), followed by Amino acids and derivatives (9%), Protein metabolism (6.68%), DNA metabolism (5.33%), Cofactors, vitamins, prosthetic groups and pigments (5.30%), Cell wall and capsule (5.19%), Respiration (4.96%), RNA metabolism (4.76%), Membrane transport (4.25%), Virulence, disease and defense (3.53%) and Stress response (3.50%) (Fig. 3).

The other subsystems detected in the *S. longiceps* gut microbiome were Nucleosides and nucleotides (3.0%), Fatty acids, lipids, and isoprenoids (2.82%), Motility and chemotaxis (2.67%), Regulation and cell signaling (2.06%), Iron acquisition and metabolism (1.42%), Nitrogen metabolism (1.24%), Phosphorus metabolism (1.21%), Potassium metabolism (1.10%), Phages, prophages, transposable elements, plasmids (0.94%), Sulfur metabolism (0.94%), Cell division and cell cycle (0.89%), Metabolism of aromatic compounds (0.36%), Secondary metabolism (0.18%), Photosynthesis (0.07%) and Dormancy and sporulation (0.06%) (Fig. 3).

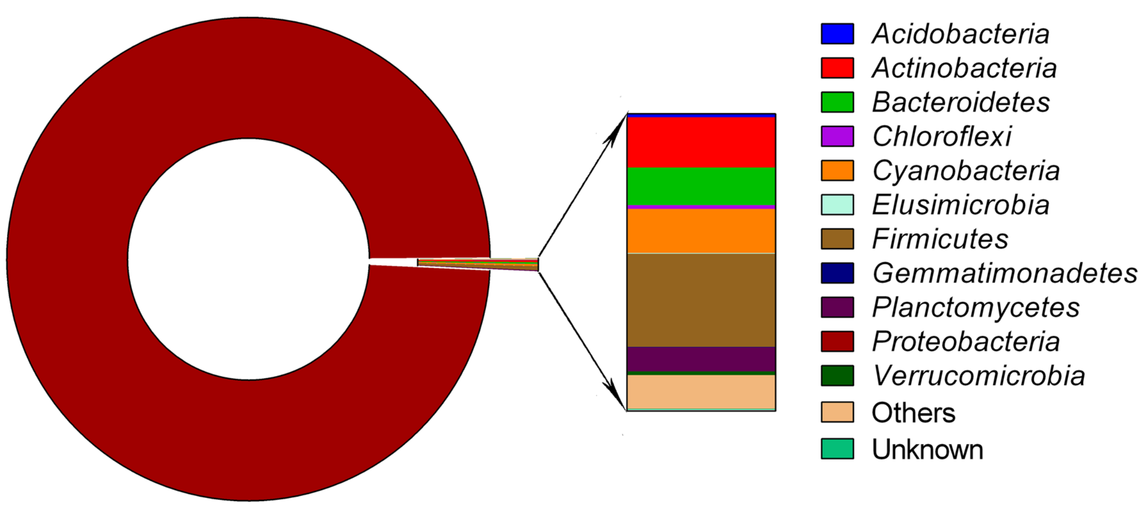

Fig. 1 Relative abundance of bacterial reads detected in *Sardinella longiceps* gut microbiome at phylum level

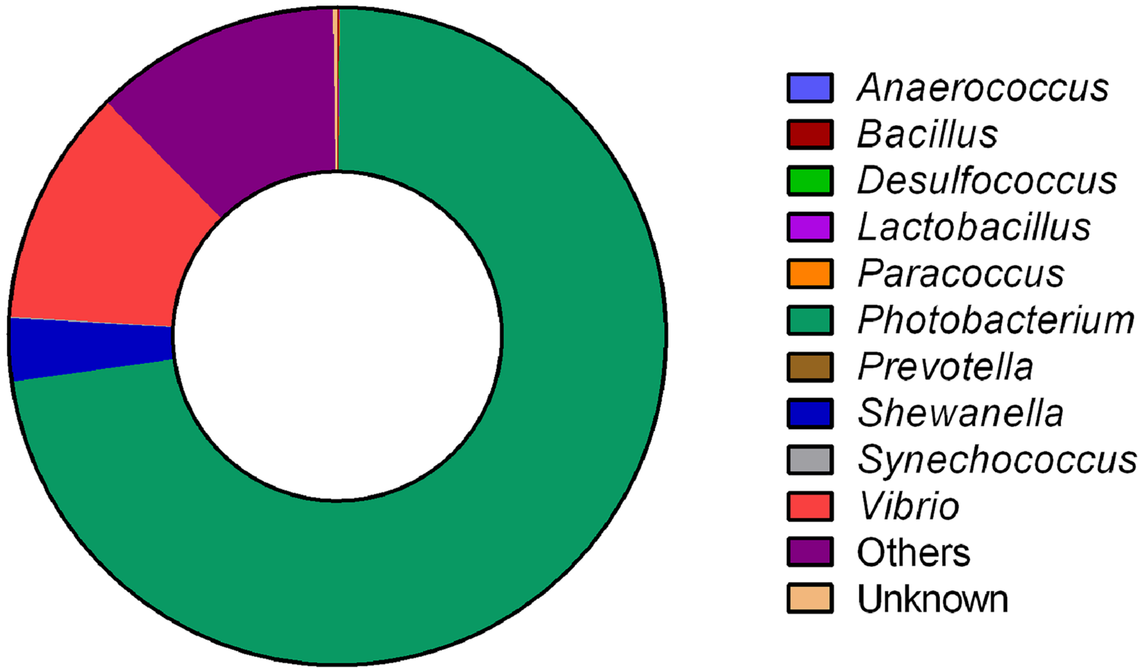

Fig. 2 Relative abundance of bacterial reads detected in *Sardinella longiceps* gut microbiome at genus level

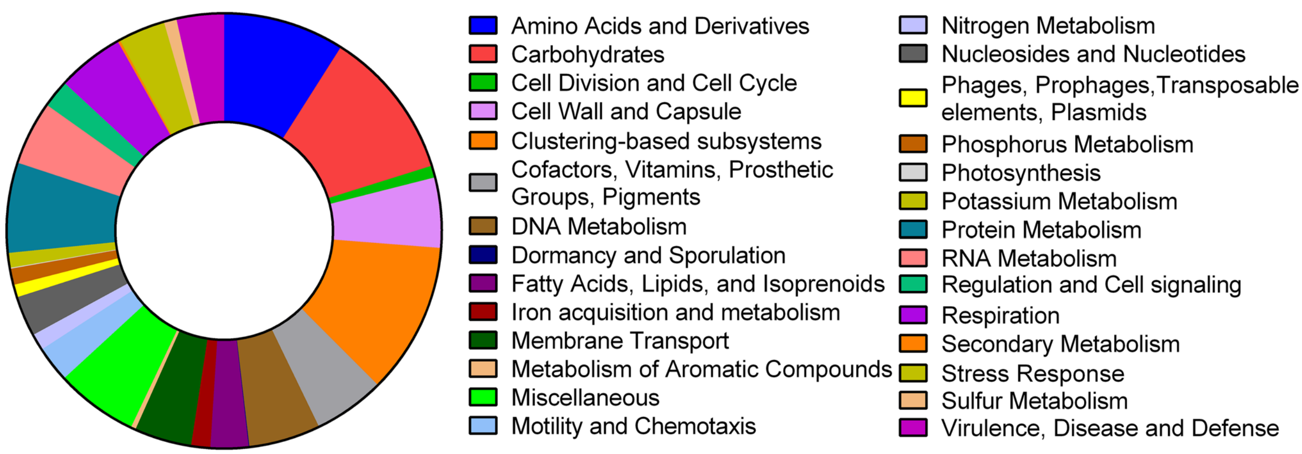

Fig.3 Relative abundance of level 1 subsystems detected in *Sardinella longiceps* gut microbiome

Resistome profiling of *Sardinella longiceps* gut microbiome

The diversity and abundance of reads related to antibiotic and heavy metal resistance were examined by a detailed analysis of the level 2 of Virulence, disease and defense subsystem. At level 2, the subsystem was dominated by Resistance to antibiotics and toxic compounds (78.99%), NULL (13.41%), Adhesion (2.37%), Invasion and intracellular resistance (1.46%), Detection (1.41%), Toxins and superantigens (1.30%) and Bacteriocins, ribosomally synthesized antibacterial peptides (1.04%) (Fig. 4). The reads related to antibiotic and heavy metal resistance falls under Resistance to antibiotics and toxic compounds subsystem and therefore, detailed analysis of the subsystems at level 3 showed an enrichment of Multidrug Resistance Efflux Pumps (37.68%), followed by Cobalt-zinc-cadmium resistance (11.60%), Resistance to fluoroquinolones (10.94%), Copper homeostasis (10.20%) and Multidrug efflux pump in *Campylobacter jejuni* (CmeABC operon) (7.16%) (Fig. 5) (Supplementary Table S3). From these results, it follows that antibiotic resistance was more pronounced than heavy metal resistance in *S. longiceps* gut microbiome.

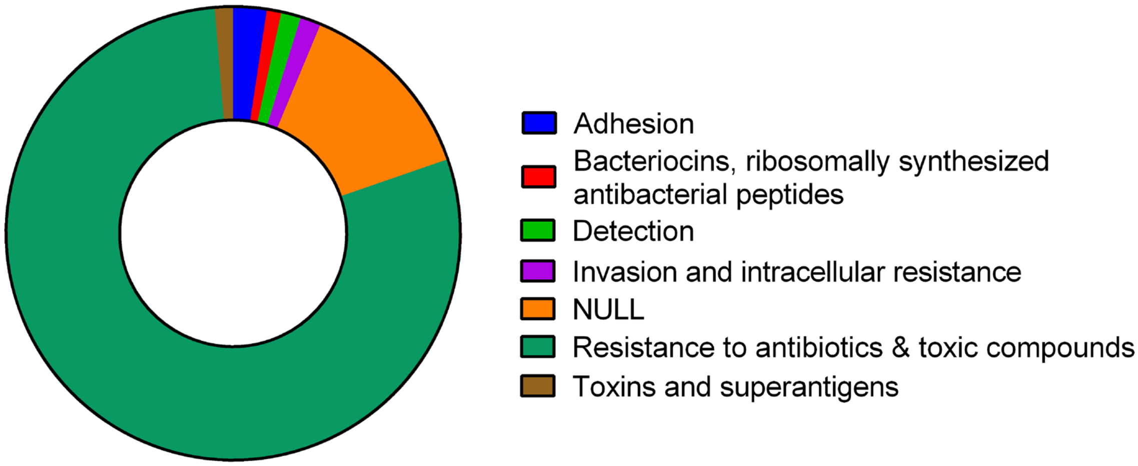

Fig. 4 Relative abundance of level 2 subsystems derived from level 1 subsystem of Virulence, disease and defense of SEED subsystems database

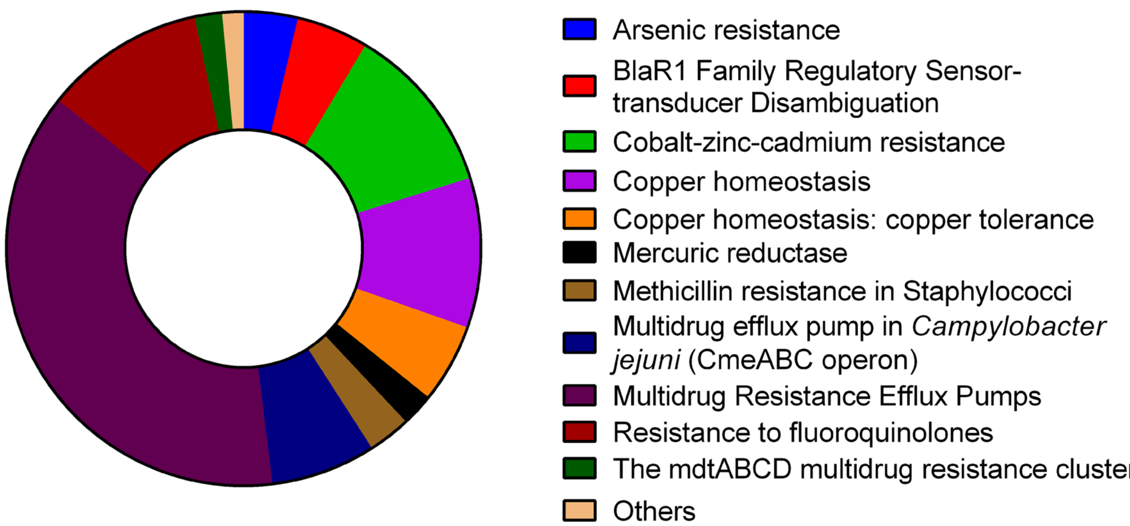

Fig. 5 Relative abundance of level 3 subsystems filtered for Resistance to antibiotics and toxic compounds

At the function level, the Resistance to antibiotics and toxic compounds subsystem was dominated by Acriflavin resistance protein (23.93%), Copper-translocating P-type ATPase (EC 3.6.3.4) (8.44%), RND efflux system, Inner membrane transporter CmeB (7.80%), Topoisomerase IV subunit A (4.76%), Multi antimicrobial extrusion protein (Na(+)/drug antiporter), MATE family of MD (4.73%) and Probable Co/Zn/Cd efflux system membrane fusion protein (4.33%) (Fig. 6) (Supplementary Table S4).

Discussion

Whole metagenome shotgun sequencing revealed that bacterial domain accounted for majority of the reads in *S. longiceps* gut microbiome (99.64%). Bacteria have been reported to be the dominant inhabitants of fish intestine and has been the sole focus of research in most fish gut microbiota studies (Rombout et al. 2011; Egerton et al. 2018). A 'core gut microbiota' concept has also been proposed with respect to fishes, indicating that similar bacterial taxa are represented in fish species from different populations as well as habitats. This implies the significance of gut bacterial communities to host functions such as digestion, nutrient assimilation and immune response (Roeselers et al. 2011).

In corroboration with our results, *Proteobacteria* and *Firmicutes* were also found to be the predominant members of the fish gut microbiota in an investigation of the gut microbiota composition of 12 finfish and 3 shark species. The relative abundance of *Proteobacteria* varied from 3 to 98% in the 15 fish species. *Firmicutes* was the next most dominant member of the gut microbiome with relative abundances

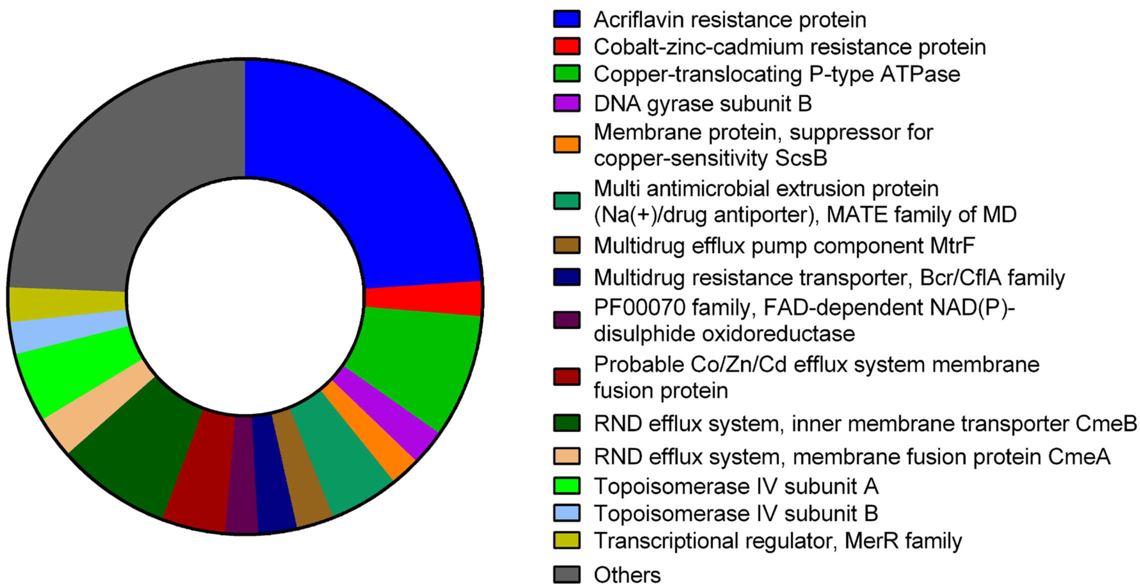

Fig. 6 Relative abundance of function level subsystems filtered for Resistance to antibiotics and toxic compounds

ranging from 1.3 to 45% (Givens et al. 2015). The high abundance of *Proteobacteria* was also observed in both clone library sequencing (99.01%) as well as pyrosequencing analyses (97.89%) of 16S rDNA amplicons of Zebra fish (*Danio rerio*) samples caught from River Shutunga, West Bengal, India (Roeselers et al. 2011). *Proteobacteria* also dominated the gut microbiota of two deep sea fishes from Japan, viz. *Chlorophthalmus albatrossis* and *Helicolenus hilgendorfi*, with relative abundances ranging from 66.4 to 98.7% (Iwatsuki et al. 2021). *Proteobacteria* contribute to significant host functions such as nutrition and that they are well adapted for survival in fish intestine and the aquatic environment (Romero et al. 2014). They comprise of autotrophs, phototrophs and heterotrophs and are well distributed in the marine environment, owing to their protean metabolic capabilities (Zhou et al. 2019).

Photobacterium sp. was found to predominate the gut microbiomes of Spinner shark (*Carcharhinus brevipinna*), Atlantic Sharpnose shark (*Rhizoprionodon terraenovae*) and Sandbar shark (*Carcharhinus plumbeus*). *Photobacterium* OTUs also constituted the core gut microflora (top 5 OTUs) of wild Mummichog, Pinfish, Silver perch, Black sea bass, Southern flounder, King mackerel, Red drum, Crevalle jack, Mahi-mahi and Great barracuda samples (Givens et al. 2015). The predominance of *Photobacterium* sp. was also established in the intestinal microbiome investigation of Atlantic cod (*Gadus morhua*) using shotgun sequencing (Riiser et al. 2019). *Photobacterium* sp. has been reported to be mutualistic bacteria of fish gut that aid in chitin digestion (Itoi et al. 2006). They are reported to possess the ability to produce polyunsaturated fatty acid- eicosapentanoic acid (EPA), antibacterial compounds and important enzymes such as asparaginases and esterases (Oku et al. 2008; Freese et al. 2009; Desriac et al. 2013; Yaacob et al. 2014; Shakiba et al. 2016; Moi et al. 2017).

Vibrio sp. has been found to have an increased representation in the gut microbiome of salt water fishes (Romero et al. 2014). They have been known to exist as symbionts in the fish gastrointestinal tract that produce hydrolytic enzymes for the breakdown of dietary ingredients. *Vibrio* sp. have been reported to produce amylase, cellulase, chitinase and lipases (Ray et al. 2012). *Shewanella* sp. along with *Vibrio* sp. are known to produce polyunsaturated fatty acids in fish gut and invertebrates (Monroig et al. 2013). The probiotic bacteria, *Bacillus* sp. showed less abundance in the *S. longiceps* gut (0.09%). This is comparable to the results from a shotgun metagenomic investigation of the freshwater carp (*Labeo rohita*) gut microbiota (Tyagi et al. 2019). *Bacillus* sp. are known to be potential contributors to fish nutrition by the production of digestive enzymes (Ray et al. 2012). The presence of *Synechococcus* (0.07%), belonging to phylum *Cyanobacteria* might be on account of the food source of *S. longiceps* that includes blue-green algae (Shah et al. 2019).

Clustering based subsystems showed the highest relative abundance in *S. longiceps* gut microbiome (11.28%). Genes are grouped under Clustering-based subsystems when their function is unknown but yet appear together in multiple genomes, indicative of functional coupling (Gerdes et al. 2011). Carbohydrates was the second most abundant subsystem in *S. longiceps* gut microbiome (11.16%). Carbohydrates are an important source of energy in animal diets. Appropriate supply and metabolism of carbohydrates is essential for normal growth, metabolism and functioning of the organism. Carbohydrate digestion provides metabolic precursors for the synthesis of other biologically important compounds (Abro 2014). Though fishes harbour enzymes for digestive processes, the gut microbiome also plays a significant role in the digestion of complex carbohydrates (Kayath et al. 2019; Butt and Volkoff 2019). The third most abundant level 1 subsystem in *S. longiceps* gut microbiome, Amino Acids and Derivatives (9%) was also found at high relative abundance in the gut microbiomes of farmed adult Turbot (Xing et al. 2013), Atlantic cod (Le Doujet et al. 2019), and Clownfish (Parris et al. 2016), implying that they constitute one of the core metabolic subsystems of gut microbiome, regardless of the host type.

The gastrointestinal microflora of fish has been suggested to enhance the digestive processes of the host by the synthesis of enzymes, cofactors and vitamins (Pérez et al. 2010). This accounts for the prevalence of Cofactors, vitamins, prosthetic groups and pigment subsystem in the *S. longiceps* gut microbiome. The subsystems related to housekeeping functions such as DNA metabolism, Respiration and RNA metabolism also constituted a major share of the annotated reads. Membrane Transport systems are vital for the modulation of bacteria-host interactions within the gut. This subsystem mapped to 4.25% of the reads at level 1 of the SEED Subsystems database. Stress Response subsystem accounted for 3.5% of the reads at the level 1 of SEED Subsystems database, implying the presence of stress response mechanisms to cope with various anthropogenic and environmental stressors present in its environment.

The level 1 subsystem of Prophages, Transposable elements, Plasmids also comprised a fraction of the annotated reads, albeit at less abundance (0.94%). These entities are most likely to favour adaptive processes in response to the changing environment. However, their presence also suggests potential risk of dissemination of antibiotic or heavy metal resistance genes between the gut bacteria by horizontal gene transfer. Such events pose a significant threat to both fish and human health. In fact, the mammalian gut has been established to be a melting pot of horizontal gene transfer (HGT) events, leading to evolutionary adaptation of the resident bacteria to the changing gut environment (Shterzer and Mizrahi 2015). Dissemination of antibiotic resistance genes via conjugal transfer was also elucidated in the intestine of a zebrafish model (Fu et al. 2017).

Virulence, disease and defense subsystem showed a moderately high relative abundance in most microbiomes. The relative abundance of this subsystem ranged between 2 and 2.5% in Australian sea lion gut (Lavery et al. 2012), piglet gut (Guevarra et al. 2018) and bovine rumen microbiomes (Singh et al. 2014), respectively. The pack animals, camel, horse, mule and donkey showed relative abundance of this subsystem at 1.7%, 2.3%, 2.6% and 3%, respectively (Dande et al. 2017). The slightly high relative abundance of this subsystem detected in *S. longiceps* gut microbiome might be an indication of contamination of the marine environment or high selective pressure existent within the gut microbiome.

Majority of the reads in Virulence, disease and defense subsystem mapped to resistance to antibiotics and toxic compounds (78.99%). The runoff of antibiotics, antibiotic resistant bacteria from terrestrial sources and production of antibiotics by the marine organisms, cause the native bacteria to evolve multifarious antibiotic resistance mechanisms (Hatosy and Martiny 2015). Marine bacteria also develop resistance mechanisms to heavy metals owing to the rampant discharge of industrial effluents into the marine environment. These heavy metals settle on the seabed or accumulate in the food chain at different trophic levels, causing the microbial community to evolve mechanisms to combat their toxicity (Ansari et al. 2004; Rahman and Singh 2018). Heavy metal contamination has also been reported to cause an increased dissemination of antimicrobial resistance on account of the overlapping genetic mechanisms for antibiotic and metal resistance. These involve co-resistance (presence of different resistance determinants on the same genetic element like plasmids, transposons or integrons), cross-resistance (single genetic determinants for both antibiotic and metal resistance) and co-regulation (shared/coordinated transcriptional and translational responses to antibiotic and metal resistance (Baker-Austin et al. 2006).

Multidrug resistance efflux pumps are important antibiotic resistance determinants in bacteria that confer resistance to different antibiotics. Apart from antibiotics, these efflux pumps also extrude other substrates such as heavy metals, dyes, detergents or organic solvents using the energy from ATP hydrolysis or proton motive force (Zgurskaya and Nikaido 2000; Blanco et al. 2016). Acriflavin resistance protein was the predominant multidrug efflux pump at the function level classification. This is a serious cause of concern as acriflavin is an important antiseptic agent with antiviral and anticancer activities owing to its topoisomerase inhibition properties (Hassan et al. 2011; Pépin et al. 2017; Tehlan et al. 2020). The widespread distribution of acriflavine resistance genes was also reported in metagenomic investigations of other anthropogenically impacted environmental samples (Guo et al. 2017; Imchen et al. 2018).

Fluoroquinolones are the first line of drugs used against nosocomial infections (Ereshefsky et al. 2017). Resistance to the broad-spectrum fluoroquinolones occur due to mutations in the bacterial DNA gyrase or topoisomerase IV that are targeted by fluoroquinolones (Redgrave et al. 2014). Both these functional categories were detected in the *S. longiceps* gut microbiome. Bacteria defend copper induced toxicity by means of P-type ATPases such as the CusCBA multiprotein complexes that pump copper out from the cells or upregulate cop operon (Nanda et al. 2019). The multidrug efflux pump CmeABC of *Campylobacter* sp. accounted for 7.16% of the reads under the level

3 subsystem of Resistance to antibiotics and toxic compounds even though *Campylobacter* sp. accounted for only 0.01% of the reads annotated against the RefSeq database.

Conclusion

Whole metagenome shotgun sequencing offered a comprehensive insight into the taxonomic and functional potential of *S. longiceps* gut microbiome. Bacteria constituted 99.64% of the sequencing reads, followed by viruses, eukaryota and archaea. At the phylum level, *Proteobacteria* comprised majority of the reads. The most abundant bacterial genera were *Photobacterium*, *Vibrio*, *Shewanella*, *Bacillus* and *Synechococcus*. Functional annotation revealed the dominance of Clustering based subsystems, followed by Carbohydrate, Amino acids and derivatives, Protein metabolism, DNA metabolism, Cofactors, vitamins, prosthetic groups and pigments and Cell wall and capsule subsystems. Notably, the gut microbiome displayed multiple mechanisms for antibiotic and heavy metal resistance, signifying positive selection of the microbial community in response to the anthropogenic perturbations on the marine ecosystem. The results from our study warrant detailed investigation of resistome profiles of other marine fish gut microbiomes to gauge the actual dimensions of this overarching threat to human health.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00203-021-02675-y.

Author contributions All the authors made substantial contributions to the design of work and drafting of the manuscript. TKJ conceived the study, performed data analysis and drafted the manuscript. RMP contributed to data acquisition, analysis and drafting of the manuscript. SGB contributed to the interpretation of data and made critical comments for the revision of the manuscript. All the authors read and approved the final manuscript.

Funding The authors gratefully acknowledge the project grants from Centre for Marine Living Resources and Ecology-Ministry of Earth Sciences, Government of India (no. MOES/10-MLR/2/2007 and MOES/10-MLRTD/03/2013) given to Dr. Sarita G. Bhat. Author Tina Kollannoor Johny acknowledges UGC, Government of India for research fellowship.

Availability of data and material The dataset generated during the current study is available in the NCBI BioSample Submission Portal as BioProject PRJNA433183 and SRA accession number SRR6677342.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

Abro R (2014) Digestion and metabolism of carbohydrates in fish. https://pub.epsilon.slu.se/10974/. Accessed 2 May 2020

Ansari TM, Marr IL, Tariq N (2004) Heavy metals in marine pollution perspective—a mini review. J Appl Sci 4:1–20. https://doi.org/10.3923/jas.2004.1.20

Augustine A, Joseph I (2018) Four novel strains of cellulolytic symbiotic bacteria isolated and characterized from GI tract of marine fishes of various feeding habits. Biocatal Agric Biotechnol 16:706–714. https://doi.org/10.1016/j.bcab.2018.05.009

Baker-Austin C, Wright MS, Stepanauskas R, McArthur JV (2006) Co-selection of antibiotic and metal resistance. Trends Microbiol 14:176–182. https://doi.org/10.1016/j.tim.2006.02.006

Blanco P, Hernando-Amado S, Reales-Calderon JA et al (2016) Bacterial multidrug efflux pumps: much more than antibiotic resistance determinants. Microorganisms 4:14. https://doi.org/10.3390/microorganisms4010014

Butt RL, Volkoff H (2019) Gut microbiota and energy homeostasis in fish. Front Endocrinol (Lausanne) 10:9. https://doi.org/10.3389/fendo.2019.00009

Dande SS, Patil NV, Joshi CG (2017) Phylogenetic and Functional diversity of faecal microbiome of pack animals. In: Abdurakhmonov IY (ed) Phylogenetics. IntechOpen, London, United Kingdom, pp 81–92. https://doi.org/10.5772/intechopen.69252

Desriac F, Jégou C, Balnois E et al (2013) Antimicrobial peptides from marine proteobacteria. Mar Drugs 11:3632–3660. https://doi.org/10.3390/md11103632

Egerton S, Culloty S, Whooley J et al (2018) The gut microbiota of marine fish. Front Microbiol 9:873. https://doi.org/10.3389/fmicb.2018.00873

Ereshefsky BJ, Al-Hasan MN, Gokun Y, Martin CA (2017) Comparison of β-lactam plus aminoglycoside versus β-lactam plus fluoroquinolone empirical therapy in serious nosocomial infections due to Gram-negative bacilli. J Chemother 29:30–37. https://doi.org/10.1080/1120009X.2016.1154684

Foster JA, Bunge J, Gilbert JA, Moore JH (2012) Measuring the microbiome: perspectives on advances in DNA-based techniques for exploring microbial life. Brief Bioinform 13:420–429. https://doi.org/10.1093/bib/bbr080

Freese E, Rütters H, Köster J et al (2009) Gammaproteobacteria as a possible source of eicosapentaenoic acid in anoxic intertidal sediments. Microb Ecol 57:444–454. https://doi.org/10.1007/s00248-008-9443-2

Fu J, Yang D, Jin M et al (2017) Aquatic animals promote antibiotic resistance gene dissemination in water via conjugation: role of different regions within the zebra fish intestinal tract, and impact on fish intestinal microbiota. Mol Ecol 26:5318–5333. https://doi.org/10.1111/mec.14255

Gerdes S, El Yacoubi B, Bailly M et al (2011) Synergistic use of plant-prokaryote comparative genomics for functional annotations. BMC Genom 12:S2. https://doi.org/10.1186/1471-2164-12-S1-S2

Ghanbari M, Kneifel W, Domig KJ (2015) A new view of the fish gut microbiome: advances from next-generation sequencing. Aquaculture 448:464–475. https://doi.org/10.1016/j.aquaculture.2015.06.033

Givens C, Ransom B, Bano N, Hollibaugh J (2015) Comparison of the gut microbiomes of 12 bony fish and 3 shark species. Mar Ecol Prog Ser 518:209–223. https://doi.org/10.3354/meps11034

Guevarra RB, Hong SH, Cho JH et al (2018) The dynamics of the piglet gut microbiome during the weaning transition in association with health and nutrition. J Anim Sci Biotechnol 9:54. https://doi.org/10.1186/s40104-018-0269-6

Guo J, Li J, Chen H et al (2017) Metagenomic analysis reveals wastewater treatment plants as hotspots of antibiotic resistance genes

and mobile genetic elements. Water Res 123:468–478. https://doi.org/10.1016/j.watres.2017.07.002
Hassan S, Laryea D, Mahteme H et al (2011) Novel activity of acriflavine against colorectal cancer tumor cells. Cancer Sci 102:2206–2213. https://doi.org/10.1111/j.1349-7006.2011.02097.x
Hatosy SM, Martiny AC (2015) The ocean as a global reservoir of antibiotic resistance genes. Appl Environ Microbiol 81:7593–7599. https://doi.org/10.1128/AEM.00736-15
Imchen M, Kumavath R, Barh D et al (2018) Comparative mangrove metagenome reveals global prevalence of heavy metals and antibiotic resistome across different ecosystems. Sci Rep 8:11187. https://doi.org/10.1038/s41598-018-29521-4
Itoi S, Okamura T, Koyama Y, Sugita H (2006) Chitinolytic bacteria in the intestinal tract of Japanese coastal fishes. Can J Microbiol 52:1158–1163. https://doi.org/10.1139/w06-082
Iwatsuki T, Kanazawa T, Ogasawara T et al (2021) 16S rRNA gene amplicon sequencing of gut microbiota in three species of deep-sea fish in Suruga Bay, Japan. Microbiol Resour Announc 10:e01260-e1320. https://doi.org/10.1128/MRA.01260-20
Johny TK, Puthusseri RM, Bhat SG (2021) A primer on metagenomics and next-generation sequencing in fish gut microbiome research. Aquac Res 52:4574–4600. https://doi.org/10.1111/are.15373
Kayath CA, IbalaZamba A, Goma-Tchimbakala J et al (2019) Microbiota landscape of gut system of Guppy Fish (*Poecilia reticulata*) plays an outstanding role in adaptation mechanisms. Int J Microbiol 2019:3590584
Khurana H, Singh DN, Singh A et al (2020) Gut microbiome of endangered *Tor putitora* (Ham.) as a reservoir of antibiotic resistance genes and pathogens associated with fish health. BMC Microbiol 20:249. https://doi.org/10.1186/s12866-020-01911-7
Kripa V, Mohamed KS, Koya KPS et al (2018) Overfishing and climate drives changes in biology and recruitment of the Indian Oil Sardine *Sardinella longiceps* in Southeastern Arabian Sea. Front Mar Sci 5:443. https://doi.org/10.3389/fmars.2018.00443
Lavery TJ, Roudnew B, Seymour J et al (2012) High nutrient transport and cycling potential revealed in the microbial metagenome of Australian Sea Lion (*Neophoca cinerea*) Faeces. PLoS ONE 7:e36478. https://doi.org/10.1371/journal.pone.0036478
Le Doujet T, De Santi C, Klemetsen T et al (2019) Closely-related *Photobacterium* strains comprise the majority of bacteria in the gut of migrating Atlantic cod (*Gadus morhua*). Microbiome 7:64. https://doi.org/10.1186/s40168-019-0681-y
Liu H, Guo X, Gooneratne R et al (2016) The gut microbiome and degradation enzyme activity of wild freshwater fishes influenced by their trophic levels. Sci Rep 6:1–12. https://doi.org/10.1038/srep24340
Moi IM, Roslan NN, Leow ATC et al (2017) The biology and the importance of *Photobacterium* species. Appl Microbiol Biotechnol 101:4371–4385. https://doi.org/10.1007/s00253-017-8300-y
Monroig Ó, Tocher DR, Navarro JC (2013) Biosynthesis of polyunsaturated fatty acids in marine invertebrates: recent advances in molecular mechanisms. Mar Drugs 11:3998–4018. https://doi.org/10.3390/md11103998
Nanda M, Kumar V, Sharma DK (2019) Multimetal tolerance mechanisms in bacteria: the resistance strategies acquired by bacteria that can be exploited to 'clean-up' heavy metal contaminants from water. Aquat Toxicol 212:1–10. https://doi.org/10.1016/j.aquatox.2019.04.011
Oku N, Kawabata K, Adachi K et al (2008) Unnarmicins A and C, new antibacterial depsipeptides produced by marine bacterium *Photobacterium* sp. MBIC06485. J Antibiot 61:11–17. https://doi.org/10.1038/ja.2008.103
Olayinka-Olagunju JO, Dosumu AA, Olatunji-Ojo AM (2021) Bioaccumulation of heavy metals in pelagic and benthic fishes of Ogbese River, Ondo State, South-Western Nigeria. Water Air Soil Pollut 232:44. https://doi.org/10.1007/s11270-021-04987-7
Parris DJ, Brooker RM, Morgan MA et al (2016) Whole gut microbiome composition of damselfish and cardinalfish before and after reef settlement. PeerJ 4:e2412. https://doi.org/10.7717/peerj.2412
Pépin G, Nejad C, Thomas BJ et al (2017) Activation of cGAS-dependent antiviral responses by DNA intercalating agents. Nucleic Acids Res 45:198–205. https://doi.org/10.1093/nar/gkw878
Pérez T, Balcázar JL, Ruiz-Zarzuela I et al (2010) Host–microbiota interactions within the fish intestinal ecosystem. Mucosal Immunol 3:355–360. https://doi.org/10.1038/mi.2010.12
Rahman Z, Singh VP (2018) Assessment of heavy metal contamination and Hg-resistant bacteria in surface water from different regions of Delhi, India. Saudi J Biol Sci 25:1687–1695. https://doi.org/10.1016/j.sjbs.2016.09.018
Ray AK, Ghosh K, Ringø E (2012) Enzyme-producing bacteria isolated from fish gut: a review. Aquac Nutr 18:465–492. https://doi.org/10.1111/j.1365-2095.2012.00943.x
Redgrave LS, Sutton SB, Webber MA, Piddock LJV (2014) Fluoroquinolone resistance: mechanisms, impact on bacteria, and role in evolutionary success. Trends Microbiol 22:438–445. https://doi.org/10.1016/j.tim.2014.04.007
Riesenfeld CS, Schloss PD, Handelsman J (2004) Metagenomics: genomic analysis of microbial communities. Annu Rev Genet 38:525–552. https://doi.org/10.1146/annurev.genet.38.072902.091216
Riiser ES, Haverkamp THA, Varadharajan S et al (2019) Switching on the light: using metagenomic shotgun sequencing to characterize the intestinal microbiome of Atlantic cod. Environ Microbiol 21:2576–2594. https://doi.org/10.1111/1462-2920.14652
Ringø E, Zhou Z, Vecino JLG et al (2016) Effect of dietary components on the gut microbiota of aquatic animals. A never-ending story? Aquac Nutr 22:219–282. https://doi.org/10.1111/anu.12346
Roeselers G, Mittge EK, Stephens WZ et al (2011) Evidence for a core gut microbiota in the zebrafish. ISME J 5:1595–1608. https://doi.org/10.1038/ismej.2011.38
Rombout JHWM, Abelli L, Picchietti S et al (2011) Teleost intestinal immunology. Fish Shellfish Immun 31:616–626. https://doi.org/10.1016/j.fsi.2010.09.001
Romero J, Ringø E, Merrifield DL (2014) The gut microbiota of fish. In: Merrifield D, Ringø E (eds) Aquaculture nutrition. Wiley, Chichester, pp 75–100
Sáenz JS, Marques TV, Barone RSC et al (2019) Oral administration of antibiotics increased the potential mobility of bacterial resistance genes in the gut of the fish *Piaractus mesopotamicus*. Microbiome 7:24. https://doi.org/10.1186/s40168-019-0632-7
Saidumohamed BE, Bhat SG (2021) Indian Oil Sardine (*Sardinella longiceps*) gut derived *Bacillus safensis* SDG14 with enhanced probiotic competence for food and feed applications. Food Res Int 150:110475. https://doi.org/10.1016/j.foodres.2021.110475
Saidumohamed BE, Baburaj AP, Johny TK et al (2021) A Magainin-2 like bacteriocin BpSl14 with anticancer action from fish gut *Bacillus safensis* SDG14. Anal Biochem 627:114261. https://doi.org/10.1016/j.ab.2021.114261
Shah N, Tang H, Doak TG, Ye Y (2011) Comparing bacterial communities inferred from 16S rRNA gene sequencing and shotgun metagenomics. Pac Symp Biocomput. https://doi.org/10.1142/9789814335058_0018
Shah TH, Chakraborty SK, Kumar T et al (2019) Food and feeding habits of oil sardine *Sardinella longiceps* from Ratnagiri coast off Maharashtra India. IJMS 48:309–318
Shakiba MH, Ali MSM, Rahman RNZRA et al (2016) Cloning, expression and characterization of a novel cold-adapted GDSL family esterase from *Photobacterium* sp. strain J15. Extremophiles 20:45–55. https://doi.org/10.1007/s00792-015-0796-4
Shterzer N, Mizrahi I (2015) The animal gut as a melting pot for horizontal gene transfer. Can J Microbiol 61:603–605. https://doi.org/10.1139/cjm-2015-0049

Singh KM, Reddy B, Patel AK et al (2014) Metagenomic analysis of buffalo rumen microbiome: Effect of roughage diet on Dormancy and Sporulation genes. Meta Gene 2:252–268. https://doi.org/10.1016/j.mgene.2014.01.005

Sullam KE, Essinger SD, Lozupone CA et al (2012) Environmental and ecological factors that shape the gut bacterial communities of fish: a meta-analysis. Mol Ecol 21:3363–3378. https://doi.org/10.1111/j.1365-294X.2012.05552.x

Susmitha V, Mathew A, Joseph I (2017) Isolation and characterization of carotenoid producing bacteria from gut of Indian oil sardine *Sardinella longiceps* (Valenciennes, 1847). In: Thomas SN, Rao BM, Madhu VR, Asha KK, Binsi PK, Viji P, Sajesh VK, Jha PN (eds) Fostering innovations in fisheries and aquaculture: focus on sustainability and safety—book of abstracts, 11th Indian Fisheries and Aquaculture Forum, ICAR-Central Institute of Fisheries Technology, Kochi and Asian Fisheries Society, Indian Branch, 21–24 November, 2017, Kochi, India, pp 441–442

Tehlan A, Karmakar BC, Paul S et al (2020) Antibacterial action of acriflavine hydrochloride for eradication of the gastric pathogen *Helicobacter pylori*. FEMS Microbiol Lett 367:178. https://doi.org/10.1093/femsle/fnaa178

Tina KJ, Bindiya ES, RaghulSubin S et al (2014) Appraisal of extraction protocols for metagenomic DNA from fish gut microbiota. IJAIR 3:7–13

Tringe SG, von Mering C, Kobayashi A et al (2005) Comparative metagenomics of microbial communities. Science (new York, N.y.) 308:554–557. https://doi.org/10.1126/science.1107851

Tyagi A, Singh B, Thammegowda NKB, Singh NK (2019) Shotgun metagenomics offers novel insights into taxonomic compositions, metabolic pathways and antibiotic resistance genes in fish gut microbiome. Arch Microbiol 201:295–303. https://doi.org/10.1007/s00203-018-1615-y

Walter JM, Bagi A, Pampanin DM (2019) Insights into the potential of the Atlantic cod gut microbiome as biomarker of oil contamination in the marine environment. Microorganisms 7:209. https://doi.org/10.3390/microorganisms7070209

Xing M, Hou Z, Yuan J et al (2013) Taxonomic and functional metagenomic profiling of gastrointestinal tract microbiome of the farmed adult turbot (*Scophthalmus maximus*). FEMS Microbiol Ecol 86:432–443. https://doi.org/10.1111/1574-6941.12174

Yaacob M, Hasan W, Ali M et al (2014) Characterisation and molecular dynamic simulations of J15 asparaginase from *Photobacterium* sp. strain J15. Acta Biochim Pol 61:745–752

Zgurskaya HI, Nikaido H (2000) Multidrug resistance mechanisms: drug efflux across two membranes. Mol Microbiol 37:219–225. https://doi.org/10.1046/j.1365-2958.2000.01926.x

Zhou Z, Yao B, Romero J et al (2014) Methodological approaches used to assess fish gastrointestinal communities. In: Merrifield D, Ringo E (eds) Aquaculture nutrition. Wiley, New York, pp 101–127

Zhou Z, Tran PQ, Kieft K, Anantharaman K (2019) Genome diversification in globally distributed novel marine *Proteobacteria* is linked to environmental adaptation. bioRxiv. https://doi.org/10.1101/814418